

Assignment 2; due Thursday, October 8, 2026

- [1] Let $(\mathbf{x}_k)_{k \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Write $\mathbf{x}_k = (x_{k1}, \dots, x_{kn})$, so that $(x_{kj})_{k \in \mathbb{N}}$ is a sequence in $\mathbb{R}$ for each $j = 1, \dots, n$.
- [a] Let $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$. Prove that $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{a}$ if and only if $\lim_{k \rightarrow \infty} x_{kj} = a_j$ for each $j = 1, \dots, n$. (*Don't forget to prove both directions!*)
- [b] Prove that $(\mathbf{x}_k)_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}^n$ if and only if $(x_{kj})_{k \in \mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$ for each $j = 1, \dots, n$. (*Don't forget to prove both directions!*)

- [2] The two parts of this problem *are* related.
- [a] Let $E \subseteq \mathbb{R}$ be nonempty. If E is bounded below, so that $\inf E$ exists, prove that there exists a sequence (a_k) with $a_k \in E$ for all k , and $\lim_{k \rightarrow \infty} a_k = \inf E$.
- [b] Let $A \subseteq \mathbb{R}^n$ be nonempty, and $\mathbf{x} \in \mathbb{R}^n$. We define the *distance* from A to $\mathbf{x}$ to be

$$\text{dist}_A(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}.$$

(Note that $\text{dist}_A(\mathbf{x})$ is well-defined and $\text{dist}_A(\mathbf{x}) \geq 0$, because it is the infimum of a nonempty set of nonnegative real numbers.) Prove that there exists a sequence $(\mathbf{y}_k)$ in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k and $\lim_{k \rightarrow \infty} \|\mathbf{x} - \mathbf{y}_k\| = \text{dist}_A(\mathbf{x})$.

- [3] Let $A \subseteq \mathbb{R}^n$. We define the *closure* of A , denoted by $\overline{A}$, to be the subset

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \emptyset \text{ for all } \varepsilon > 0\}.$$

That is, a point $\mathbf{x} \in \mathbb{R}^n$ lies in the closure of A if and only if every open ball of any positive radius intersects A .

- [a] Show that $A \subseteq \overline{A}$.
- [b] Show that $\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \emptyset \text{ for any open set } U \text{ containing } \mathbf{x}\}$.
- [c] Show that $\overline{A}$ is closed.
- Hint:* First show there exists ε_0 such that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \emptyset$, and then show that $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \emptyset$. You will need to use part [b].
- [d] If A is closed, show that $\overline{A} = A$.
- (Combining parts [c] and [d] shows that A is closed if and only if $\overline{A} = A$.)

- [4] Let $E \subseteq \mathbb{R}^n$ be nonempty.
- [a] If $(\mathbf{x}_k)_{k \in \mathbb{N}}$ is a sequence such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$, prove that $\mathbf{p} \in \overline{E}$.
- [b] Let $\mathbf{p} \in \overline{E}$. Prove that there exists a sequence $(\mathbf{x}_k)_{k \in \mathbb{N}}$ such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k \rightarrow \infty} \mathbf{x}_k = \mathbf{p}$. *Hint:* Consider $B_{\frac{1}{k}}(\mathbf{p})$.

Taken together, these two statements say that $\overline{E}$ is exactly the set of limits of convergent sequences whose elements all lie in E .

- [5] Let $(\mathbf{x}_m)_{m \in \mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Define a new sequence $(\mathbf{s}_k)_{k \in \mathbb{N}}$ by

$$\mathbf{s}_k = \sum_{m=1}^k \mathbf{x}_m.$$

If the sequence $(\mathbf{s}_k)$ converges, then we say that *the series* $\sum_{m=1}^{\infty} \mathbf{x}_m$ *converges*. If $\lim_{k \rightarrow \infty} \mathbf{s}_k = \mathbf{s}$, then we write $\sum_{m=1}^{\infty} \mathbf{x}_m = \mathbf{s}$.

- [a] Prove that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges iff: for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon$$

for all $\ell \geq k \geq N$.

Hint: A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy.

- [b] Suppose that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges. For any $N \in \mathbb{N}$, prove that $\sum_{m=N}^{\infty} \mathbf{x}_m$ converges, and that

$$\lim_{N \rightarrow \infty} \left(\sum_{m=N}^{\infty} \mathbf{x}_m \right) = 0.$$

Hint: Consider the sequence $(\tilde{\mathbf{s}}_{\ell})_{\ell \geq N}$ given by $\tilde{\mathbf{s}}_{\ell} = \sum_{m=N}^{\ell} \mathbf{x}_m$.

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